MILITARY SPECIFICATION

GENERATOR SYSTEM; ALTERNATOR
RECTIFIER 30/80 AMPERES, 28 VOLTS

1. SCOPE

1.1 Scope. This specification covers one type electrical generator system, consisting of an alternator, solid state-rectifier assembly and solid state voltage regulator in one housing. The system is belt driven and delivers a rated output of 30 amperes when operated in air and 80 amperes when submerged in water.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS
Military

MIL-E-5272	-	Environmental Testing, Aeronautical and Associated Equipment, General Specification for.
MIL-B-11188	-	Batteries, Storage: Lead-Acid.
MIL-C-13927	-	Fungus Resistance Test; Automotive Components.
MIL-E-55301	-	Electromagnetic Compatibility.

STANDARDS
Military

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	-	Identification Marking of US Military Property.
MIL-STD-202	-	Test Methods for Electronic and Electrical Component Parts.

DRAWINGS

11613630	-	Rectified AC Generating System.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402).

3. REQUIREMENTS

3.1 Preproduction sample. Preproduction samples of the generator systems to be furnished under this specification shall be produced prior to the manufacture of these items in production quantity. The preproduction samples shall be submitted to the Government for preproduction inspection to determine conformance to the quality assurance provisions of this specification (see 4.2). The preproduction samples shall be fully representative of generator systems to be supplied from production facilities and tooling.

3.2 Materials. Materials shall be as specified herein, in applicable specifications and on applicable drawings. Materials not specifically designated shall be of good commercial quality. Materials shall be suitable for use in the generator system operating over an ambient temperature range from minus 65°F to plus 180°F without any change in physical or dimensional properties that would result in calibration or operation of the generator system to fall outside of the limits specified herein.

3.3 Design and construction. Generator systems fabricated in accordance with this specification shall have external dimensions that fall within the envelope outline of Drawing 11613630. Electrical connector mounting and shaft dimensions and location shall conform with the detail requirements of Drawing 11613630.

3.4 Polarity. The generator system shall be designed to operate with a negatively-grounded electrical circuit.

3.5 Standard parts. Military standard parts shall be used whenever possible.

3.6 Threaded surfaces. All threaded surfaces shall be in the inch system and shall conform to National Bureau of Standards Handbook H28.

3.7 Dissimilar metals. Except where necessary to compplete an electrical circuit, contact between dissimilar metals, which would encourage galvanic action, shall be avoided. Separation of dissimilar metals shall be accomplished by providing insulation between mating surfaces.

3.8 Performance

3.8.1 Rated output air. When tested as specified in 4.5.2, the generator system shall meet the performance requirements specified for rated output in air on Drawing 1161360.

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3.8.1.1 Two minute output test, air. When tested in accordance with 4.5.2.1, the generator system shall deliver a minimum output of 30 amperes at not more than 2000 RPM.

3.8.2 Rated output, submerged. When tested as specified in 4.5.3, the generator system shall meet the performance requirements specified for submerged rated output on Drawing 11613630. The SW4 current limiting switch must be open when running submerged.

3.8.3 Voltage regulation. When tested as specified in 4.5.2, 4.5.2.1, 4.5.3 and 4.5.4, the output voltage of the generator system shall be established and maintained at set voltage (see 6.4.1) ± 0.5 volts under the load and speed conditions indicated. In addition, the generator system shall reestablish and maintain the output voltage at set voltage ± 0.5 volts within 0.6 seconds of any load change.

3.8.3.1 Voltage regulation quality conformance test. When tested in accordance with 4.5.4.1, the output voltage shall be set voltage ± 0.5 within 0.6 seconds of any load change.

3.8.4 Voltage characteristics.

3.8.4.1 Voltage transients. When tested as specified in 4.5.5.1, the generator system shall withstand the transient energy pulse generated without physical damage or impairment of performance. After the test, the generator system shall meet the rated output performance at the 30 ampere (amp) point.

3.8.4.2 Voltage adjustment when tested as specified in 4.5.5.2. Voltage adjustment shall be such as to provide a set voltage over the minimum range from 26 to 30 volts.

(a) The voltage adjustment rheostat shall withstand the test specified in 4.5.5.2 without developing erratic wiper contact.
(b) In adjusting the rheostat over its full range, the current shall be continuously variable and be no less than the normal minimum obtained at maximum rheostat resistance.

3.8.4.3 Ignition lead current. When tested as specified in 4.5.5.3, the current flowing through the ignition switch circuit shall not exceed 300 milliamperes and with the ignition circuit open, the current flowing from the battery shall not exceed 25 milliamperes.

3.8.5 Endurance. After having been subjected to the test specified in 4.5.6, the generator system shall meet the requirements of 3.8.1 and 3.8.3. There shall be no servicing, adjustment or replacement of parts.

3.8.6 Overspeed. After having been subjected to the test specified in 4.5.7, the generator system shall meet the rated output performance at the 30 AMP point. At the conclusion of the test, the generator system shall be visually examined. The system is required to withstand this test without any evidence of electrical or mechanical damage.

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3.8.7 High temperature. After having been subjected to the test specified in 4.5.8 for 200 hours, the generator system shall meet the rated output performance at the 30 AMP point. During the test, voltage regulation shall conform to 3.8.3, except that when 30 AMPS cannot be obtained, the maximum load that can be obtained at each speed before regulator stops switching shall be used.

3.8.8 Low temperature. After having been subjected to the test specified in 4.5.9, the generator system shall meet the rated output performance at the 30 AMP point. During the test, voltage regulation shall conform to 3.8.3.

3.8.9 Shock. After having been subjected to the test specified in 4.5.10, the generator system shall meet the rated output performance at the 30 AMP point. During the 30 AMP rated output test, voltage regulation shall conform to 3.8.3.

3.8.10 Vibration. After having been subjected to the test specified in 4.5.11, the generator system shall meet the rated output performance, at the 30 AMP point. After the test the voltage regulation shall conform to 3.8.3.

3.8.11 Submersion. After having been subjected to the test specified in 4.5.12, the generator system shall meet the rated output performance at the 30 AMP point. During the 30 AMP rated output test the voltage shall remain at set voltage $\pm$ 0.5 volts. At the conclusion of the test, there shall be not more than 1 cubic centimeter of salt water in the regulator compartment.

3.8.12 Fungus resistance. After having been subjected to the test specified in 4.5.13, the generator system shall evidence no microbial growth that would adversely affect performance and shall meet the rated output performance at the 30 AMP point. During the 30 AMP rated output test voltage regulation shall conform to 3.8.3.

3.8.13 Corrosion resistance. After having been subjected to the test specified in 4.5.14, the generator system shall meet the rated output performance at the 30 AMP point. During the 30 AMP output test, voltage regulation shall conform to 3.8.3.

3.8.14 Electromagnetic compatibility. When tested as specified in 4.5.15, the generator system shall conform to the electromagnetic compatibility, emission and susceptibility requirements of MIL-E-55301 for tactical equipment (1.5 to 1000 megacycles).

3.8.15 Generator drive, class A only.

3.8.15.1 100-Hour. When tested as specified in 4.5.16.1, the flexible drive shall limit the armature amplitude to within plus or minus 5 degrees.

3.8.15.2 15-Minute. When tested as specified in 4.5.16.2, the flexible drive shall limit the armature amplitude to within plus or minus 7 degrees.

3.8.16 Sand and dust. After testing as specified in 4.5.17 the generator system shall duplicate the rated output performance at the 30 AMP point. During the 30 AMP test voltage regulation shall conform to 3.8.3.

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3.8.17 Road test. The generator system shall be subjected to the test specified in 4.5.18. Throughout the test, the generator system shall provide satisfactory performance without maintenance or repair. Subsequent to the test the generator system shall be inspected and tested in conformance with 3.8.1.

3.8.18 Composite. A generator system shall withstand the sequence of tests specified in 4.5.19. At the conclusion of the tests, the generator system shall meet the requirements of 3.8.1.

3.9 Painting. All exterior surfaces of the generator system except electrical connections and the extended shaft shall be painted in conformance with requirements on the applicable drawing.

3.10 Identification of product.

3.10.1 Marking. The generator system shall be marked in accordance with MIL-STD-130.

3.10.2 Nameplate. The generator system shall have a nameplate attached to the generator frame in a conspicuous location, so that plate may be easily read when generator system is installed in vehicle. The nameplate shall contain the following minimum information:

GENERATOR SYSTEM DC
30 AMP AIR 80 AMP SUMBERGED - 28 VOLTS
FEDERAL STOCK NO. (FSN)
MAXIMUM RATED SPEED - 10,000 RPM
US
SERIAL NUMBER
MANUFACTURER'S NAME OR IDENTIFICATION CODE

3.11 Workmanship. Workmanship shall be such as to assure a product free of burrs, scratches, sharp edges, chips, corrosion, scale and dirt.

4. *QUALITY ASSURANCE PROVISIONS*

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction inspection. A preproduction sample (see 3.1) of eight generator systems of each type included in the contract or purchase order shall be submitted for preporduction inspection. Preproduction inspection shall be performed by the Government at a location designated by the Government (see 6.2) and shall consist of examination for the defects specified in table II and performance tests as specified in table I in order indicated.

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Table I. Preproduction test sequence

Test	Paragraph No.	Specimen 1	2	3	4	5	6	7	8
Rated output, air	4.5.2	X	X	X	X	X			
Voltage regulation	4.5.4	X	X	X	X	X			
Rated output, submerged	4.5.3	X							
Ignition lead current	4.5.5.3	X							
Endurance	4.5.6	X							
High temperature	4.5.8		X						
Low temperature	4.5.9		X						
Shock	4.5.10		X						
Overspeed	4.5.7			X					
Vibration	4.5.11			X					
Fungus resistance	4.5.13			X					
Sand and dust	4.5.17			X					
Submersion	4.5.12				X				
Voltage transients	4.5.5.1					X			
Electromagnetic compatibility	4.5.15					X			
Generator drive (class A only)	4.5.16					X			
Composite	4.5.19							X	
Corrosion resistance	4.5.14						X	X	
Road test	4.5.18								X

4.2.1 Failure. Failure of any specimen to pass any examination or test specified herein may be cause, at the option of the Government, for refusal to conduct additional testing until the faults revealed by the tests have been corrected.

4.3 Quality conformance inspection.

4.3.1 Sampling.

4.3.1.1 Lot formation. A lot shall consist of all generator system assemblies of one type, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.3.1.2 Sampling for examination. Samples for acceptance examinations shall be selected in accordance with MIL-STD-105.

4.3.2 Examination.

4.3.2.1 Acceptable quality level. Each generator system assembly selected in accordance with 4.3.1.2 shall be examined for conformance to the following acceptable quality levels (AQL) on the basis of percent defective:

Classification	AQL
Major	1.0
Minor	2.5

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4.3.2.2 Classification of defects. For examination purposes, defects shall be as classified in table II.

Table II. Classification of defects

Categories	Defects
Major:	
101	Dimensions affecting interchangeability not within tolerance.
102	Inadequate electromagnetic compatibility devices.
Minor:	
201	Dimensions not affecting interchangeability not within tolerance.
202	Improper finish.
203	Improper marking.
204	Poor workmanship.

4.3.3 Testing inspection.

4.3.3.1 Quality conformance 100%. Generator frame temperatures shall not be stabilized, and all generators shall be subjected to the tests specified in table III.

Table III. Order of quality conformance tests

Paragraph No.	Test
4.5.2.1	Two minute output test, air
4.5.4.1	Voltage regulation quality conformance test
4.5.5.3	Ignition lead current

4.3.3.1.1 Rejection and retest. Generators that have been rejected may be reworked or have parts replaced to correct defects and be resubmitted for examination and tests. Before resubmitting, detailed information or data concerning the rejection and the action taken to correct the defects shall be furnished to the Government inspector.

4.4 Control tests.

4.4.1 Sampling for control tests. Control test samples shall be selected at the rate of one generator systems for each 250 produced except that not more than two shall be selected in any 30 day period. Samples shall be from a production lot which has passed the quality conformance examination specified in 4.3.2.2 and quality conformance tests specified in 4.3.3.1.

4.4.2 Procedure. Control test samples shall be identified as to production period and subjected to the tests specified in table IV.

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Table IV. Control tests in sequence

Test	Paragraph No.
Overspeed	4.5.7
Voltage regulation	4.5.4
Voltage transients	4.5.5.1 (b) only
Rated output, submerged	4.5.3 at 3000 RPM only
Submersion	4.5.12.2

4.4.3 Failure. Failure of a control test sample unit to pass any specified examination or test may be cause, at the option of the Government, to refuse to accept subsequent lots until it has been proved to the satisfaction of the Government that the faults revealed by the examination or tests have been corrected.

4.5 Test procedures.

4.5.1 General.

4.5.1.1 Conditions. Unless otherwise specified herein test shall be conducted in ambient air under general laboratory conditions acceptable to the Government. During the tests the generator shall be cooled by an ambient air flow at the rate of 1000 ± 50 feet per minute (FPM). Air flow shall be in the axial direction and over the periphery of the generator system. The ambient room temperature shall be maintained at 77° ± 15°F. Readings shall be corrected for variations in the ambient temperature by use of the following formula:

$$\text{Output (corrected)} = \frac{\text{Output (observed)}}{1 \pm .00214\ (77-Ta)}$$

Where Ta = ambient temperature °F.

4.5.1.2 Apparatus. In addition to standard measuring equipment (see table V, figure 1 and applicable test specifications), test appartus shall include chambers for conducting low temperature (80°F), high temperature (± 90°F) and sand and dust tests.

4.5.1.3 Calibration of test equipment. Unless otherwise specified, test equipment accuracy shall be such as to permit measurement of 10 percent of product or test specification tolerance. Records of calibration of test equipment at intervals sufficient to establish required accuracy shall be made available to the Government. The inspector may refuse to conduct an inspection, where accuracy of test equipment has not been established to the satisfaction of the Government. Standard measuring instrument tolerance accuracies are listed in tables V and VI.

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Table V. Measuring instrument tolerances for preproduction tests

Measuring instrument	Accuracy
Voltmeter DC	± 0.25 percent (30 volts - full scale)
Ammeter DC	± 0.50 percent (100 amps - full scale)
Tachometer	± 1 percent
Prony Brake	± 2 percent
Temperature indicating system	± 8°F
Milliammeter	± 0.50 percent (500 milliamps - full scale)

Table VI. Measuring instrument tolerances for quality conformance and control tests

Measuring instrument	Accuracy
Voltmeter DC	± 0.15 volts
Ammeter DC	± 0.5 Amps
Tachometer	± 50 RPM at 1800 RPM ± 200 RPM at 10,000 RPM
Temperature indicating system	± 0.8°F
Milliammeter	± 5.0 milliamps

4.5.1.4 Test circuit batteries. Batteries used in connection with tests shall conform to MIL-B-11188. Test circuit batteries shall be considered fully charged when batteries after having been stored at the ambient laboratory temperature will accept a charging current of not more than 4.5 AMPS at a set voltage. Two 12-volt batteries connected in series shall be used. In case of a high voltage reading, battery specific gravity shall be checked, and if over 1300, the battery shall be discharged until the specific gravity is less than 1300 and the charging current does not exceed 4.5. AMPS.

4.5.1.5 Frame temperature. Frame temperature shall be measured near the center of the stator lamination stack on the outside diameter.

4.5.1.6 Stabilized temperature. The generator system shall be considered to be temperature stabilized, when operation under constant conditions fails to cause a change in frame temperature in excess of 2°F in any five minute period during a 20 minute test.

4.5.1.7 Voltage setting. For all tests unless otherwise specified, the voltage regulator shall be adjusted to set voltage (28 ± 0.2 volts), with the alternator running at 1500 ± 50 revolutions per minute (RPM) and producing 10 ± 1 amps. This shall be accomplished after the frame temperature has been stabilized.

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4.5.2 Rated Output, Air. Unless otherwise specified on applicable drawing, in performing output tests in air on the generator system, the regulator voltage sensing leads shall be connected to the same point at which the voltage is measured. To determine conformance to 3.8.1, and 3.8.3 the following procedures shall be observed:

(a) Connect the generator system in a circuit as shown in figure 1.
(b) Unless otherwise specified herein, the battery shall remain connected for all tests and switches SW3 and SW4 (current limiting switch) shall be closed.
(c) Speed, frame temperature, ignition lead current, output current and system voltage shall be recorded.
(d) The voltage regulator shall be adjusted to set voltage (28 ± 0.2 volts), with the alternator running at 1500 ± 50 (RPM) and producing 10 ± 1.0 AMPS. This shall be accomplished after the frame temperature has been stabilized.
(e) The current limiting rheostat shall be adjusted to between 29.5 to 30.5 AMPS with the generator system running at 5000 ± 200 RPM With the speed immediately reduced to 2000 RPM, the current limit shall remain within the limits between 29.5 and 30.5 AMPS. The speed shall then be increased to 10,000 ± 100 RPM and the current limit must remain between 29.5 and 30.5 AMPS.
(f) The generator system speed shall be adjusted to 1500 ± 50 RPM and 1100 ± 50 RPM. The load shall be 29.0 AMPS. minimum at 1500 RPM, and 8 AMPS minimum at 1100 RPM.

4.5.2.1 Two minute output test, air. To determine conformance to 3.8.1.1, and 3.8.3 the unstabilized generator system without air flow shall be connected into the circuit in figure 1 (with the battery in the circuit and switch SW4 closed), and then subjected to the following test procedure.

(a) The voltage regulator shall be adjusted to 28 ± 0.2 volts with the generator system running at 1500 ± 50 RPM, with a load not more than 10 AMPS.
(b) The current limiter switch shall be adjusted to within the limits between 29.5 and 30.5 AMPS with the generator system running at 5000 ± 200 RPM with the speed immediately reduced to 2000 ± 100 RPM, the current limit shall remain within 0.5 AMPS of the value to which it was adjusted.
(c) Immediately after adjusting current limit point, switch SW4 (current limiting shall be opened. The minimum output shall be 55 AMPS in not less than two minutes after opening SW4.)

4.5.3 Rated output submerged. To determine conformance to 3.8.2, and 3.8.3 the generator system with SW4 open and without air shall be subjected to the output test procedure of 4.5.2 while submerged in water. During submersion the generator system shall be connected as shown on Drawing 11613630 for submerged operations. Output readings shall be taken at speeds of 1500 ± 50, 3000 ± 100 and 6000 ± 200 RPM. Loads shall be 4.5 AMPS minimum at 1500 RPM, 71 AMPS minimum at 3000 RPM and 79 AMPS minimum at 6000 RPM.

4.5.4 Voltage regulation: To determine conformance to 3.8.3, voltages shall be recorded while, the generator system is operated with SW4 closed, under the following conditions:

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(a)

RPM	AMPS
2000 ± 100	0 -4.5
2000 ± 100	30 ± 1 AMPS
2000 ± 100	0 -4.5
2000 ± 100	15 ± .50 AMPS
10,000 ± 200	Same sequence as for 2000 RPM

Change in load resistance shall be accomplished in 10 milliseconds maximum time. Total time for the test shall not exceed 10 minutes.

4.5.4.1 Voltage regulation quality conformance test. To determine conformance to 3.8.3.1, immediately after the two minute output test (4.5.2.1) the generator system shall be run for a maximum of 5 minutes without air flow. The voltage shall then be recorded with the unstabilized generator system operating in an ambient temperature of 77°F under the following conditions.

2000 ± 50	0- 4.5
2000 ± 50	30 ± 1 AMPS, 5 minutes maximum
2000 ± 50	0 - 4.5
2000 ± 50	15 ± .5 AMPS, 5 minutes maximum

4.5.5 Voltage regulator characteristics.

4.5.5.1 Voltage, transients. To determine conformance to 3.8.4.1 generator system operation at 6,000 RPM at full load (30 AMPS) connected as in Figure 1, the switch SW1 and SW4 closed and switch SW2 open, switch SW3 shall be opened and closed at a rate of 10 times per minute. Switch SW3 shall be closed for approximately five seconds, and open for one second during each cycle. The test will be conducted for 2 1/2 hours at each of the following ambient temperatures.

(a) - 65° ± 5°F
(b) ± 77° ± 15°F

Total test duration shall be 5 hours.

4.5.5.2 Voltage adjustment. Conformance of the generator system to 3.8.4.2 shall be determined as follows:

(a) A carbon zinc 1.5 volt, size "D", dry cell in series with a suitable resistor, such that the maximum current carried in the application will be carried during the test, and a milliam meter shall be applied between the center tap and one end of the rheostat for a period of four hours with rheostat at an ambient temperature of 300°F.

NOTE: The rheostat need not be installed in the generator system to conduct this test, and need not be the rheostat in alternator in (b).

(b) The alternator shall then be run at room temperature at 2500 ± 50 RPM, with a 16 AMP maximum lead. The voltage adjustment range under this condition shall be as specified (see 3.8.4.2).

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4.5.5.3 Ignition Lead Current. To determine conformance to 3.8.4.3, the current flowing through the ignition switch circuit with switches SW1, SW3, and SW4 closed and switch SW2 open (see figure 1) shall be measured.

4.5.6 Endurance. To determine conformance to 3.8.5, the generator system shall be subjected to the following tests conducted at room temperature:

(a) Set regulator at 28 ± 0.2 volts (set voltage)
(b) Adjust load for maximum rated output per 4.5.2.
(c) Operate the generator system for a total of 2000 hours in accordance with speed and duration indicated in table VI recording frame temperature, ambient temperature generator system voltage, and generator system current.
(d) During the test, switches SW4 shall be opened every 30 minutes for 10 seconds.

Table VI. Endurance sequence

RPM	Duration	Generator system position
2500	1500 hours	Shaft Horizontal
10000	500 hours	Shaft Horizontal *

NOTE: Throughout the endurance test, the belt loading shall be maintained in accordance with the formula on the applicable drawing. *Throughout this test period the load shall be cycled from between 0 - 5 AMPS to maximum rated output per 4.5.2 (30 minutes at each load).

4.5.7 Overspeed. To determine conformance to 3.8.6, the generator system shall be operated at 10000 ± 200 RPM at maximum rated output per 4.5.2 until stabilized. After stabilization (see 4.5.1.6) the load shall be reduced to less than five AMPS and the speed increased to 12000 ± 200 RPM. The 12000 ± 200 RPM speed and the corresponding load shall be continued for 15 seconds.

4.5.8 High temperature. To determine conformance to 3.8.7, the generator system shall be operated at the maximum load possible with SW4 closed at 1750 ± 50 RPM and 28 ± 0.5 volts until the frame temperatures is stabilized. This test shall be conducted at an ambient air temperature of 100°F. Air movement in the high temperature chamber shall be per 4.5.1.1.

4.5.9 Low temperature. To determine conformance to 3.8.8, the generator system shall be subjected to the following test:

(a) Prior to testing, the generator system shall be stored at a temperature of -80°F for a period of 24 hours.
(b) After removal from storage the generator system shall be placed and stabilized in a chamber with an ambient temperature of -65°F.
(c) The generator system shall be operated at 1750 ± 50 RPM with a 30 AMP load for a period of four hours.
(d) During the test, the voltage shall remain at set voltage ± 0.5 volts.

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4.5.10 Shock. To determine conformance to 3.8.9, the generator system shall be tested in accordance with test condition C, method 205, medium impact shock test of MIL-STD-202. Test procedure involving oscilloscope monitoring of the shock impulse may be used providing that the above minimum requirements are met and the method is approved by the Government.

4.5.11 Vibration. To determine conformance to 3.8.10, the generator system shall be tested in accordance with test condition D of method 204 of MIL-STD-202. The following special conditions shall apply:

(a) The frequency range shall be 10 to 600 cycles per second.
(b) The maximum load along the longitudinal (shaft) axis shall be 10G applied at the vibration tables.
(c) The generator shaft shall be rotated throughout the vibration test at a speed of 1750 ± 50 RPM, using a flexible figure 8 coupling.
(d) The generator shall be rigidly mounted as in normal vehicle operating position.

4.5.12 Submersion.

4.5..2.1 Preproduction. To determine conformance to 3.8.11, the generator system shall be stabilized (see 4.5.1.6) by operating at 28 ± 0.5 volts, 2000 ± 100 RPM and a load current of 30 AMPS. Stabilization shall be accomplished in the dry condition. The generator system shall then be subjected to the following test cycle:

(a) The generator system shall be driven in the dry condition for a period of 60 hours at 6000 ± 200 RPM and no load.
(b) The generator shaft shall then be rotated for a period of one hour while completely submerged in a 5 percent (by weight) salt solution at a generator shaft speed of 1750 ± 50 RPM, 40 AMP load.
(c) The above cycle shall be performed a total of eight times.

4.5.12.2 Control test. To determine conformance to 3.8.11, the generator system shall be stabilized (see 4.5.1.6) by operating at 28 volts, 1750 ± 50 PM, and 30 AMPS. Stabilization shall be accomplished in the dry condition. The generator shaft shall be rotated for a period of eight hours while completely submerged in a 5 percent (by weight) salt solution at a speed of 1750 ± 50 RPM, 40 AMP load.

4.5.13 Fungus resistance. To determine conformance to 3.8.12, the generator system shall be tested in accordance with class 2, method B of MIL-F-13927 except that the test period shall be a continous 90 days. Upon removal from the cabinet, examination shall be accomplished by means of removable plates or plugs, without disassembly and without disturbing any fungus growths that have developed. Upon completion of the examination and replacement of access port covers, the generator system shall be operated for one hour, without load at 6000 ± 200 RPM before being subjuected to the additional testing specified in 3.8.12.

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4.5.14 Corrosion resistance. To determine conformance to 3.8.13, the generator system shall be tested for resistance to corrosion in accordance with method 101 of MIL-STD-202, except that the length of the test shall be 200 hours. During the test the generator system shaft will be rotated for a ten minute period out of each 8 hour test interval. After the test the generator system shaft shall be rotated for 10 minutes at 1750 ± 50 RPM and no load before being subjected to the additional testing specified in 3.8.13.

4.5.15 Electromagnetic compatibility. To determine conformance to 3.8.14, the generator system shall be tested in accordance with the applicable provisions of MIL-E-55301.

4.5.16 Generator drive, class A only. To determine conformance to 3.8.15, class A generators shall be tested as specified in 4.5.16.1 and 4.5.16.2. The test shall be conducted on a universal joint, torsional-vibration machine which has a flywheel with a moment of inertia at least twenty times that of the rotor being tested. Critical frequencies shall be determined by a scanning search through the applicable range and amplitude of torsional vibration.

4.5.16.1 100-Hour. The rotor shall be subjected to a 100-hour torsional vibration with plus or minus one degree torsional amplitude input to the drive shaft at critical frequencies.

4.5.16.2 15-Minute. The rotor shall be subjected to a 15-minute torsional vibration with plus or minus two degrees input to the drive shaft at critical frequencies.

4.5.17 Sand and dust. To determine conformance to 3.8.16, the generator shall be subject to the procedure 1 sand and dust test of MIL-E-5272, the generator position in the dust chamber shall be as in intended use. The dust velocity shall be 2,300 ± 500 feet per minute. For 10 minutes of each hour of the test period, the generator shall be driven at a speed of 3450 ± RPM.

4.5.18 Road test. To determine conformance to 3.8.17, two generator system test samples shall be subjected to the following test procedure, conducted at Government proving grounds.

(a) The two generator test samples shall be installed in two M656, 5 ton trucks. Prior to the procedure indicated below, the vehicles shall have been subjected to a 500 mile break-in run to provide preliminary assurance that each vehicle and all components thereof are properly assembled and adjusted, and in proper working order. Thereafter, the vehicle shall be subjected to the test procedure of table VII.

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Table VII. 20,000 Mile Test

Course	Mileage & speeds	Vehicle payload
Hard surface	8,000 miles paved at verying up to maximum	Highway
Secondary roads	6,000 miles at speeds applicable to conditions of terrain	Cross-country
Hilly cross-country	6,000 miles at speeds applicable to conditions of terrain	Cross-country
Swimming	Every 2500 miles during this test	Rated payload

NOTE: Fifty percent of the test of the M656 vehicle only shall be performed while towing loads specified in 6.3.

4.5.19 Composite. To determine conformance to 3.8.18, one sample generator system shall be subjected to the following cycle of laboratory tests, except that operation at the 30 AMP point to check the effect of the tests, shall be omitted in items (1) thru (14).

(1) 4.5.2 - Rated output, air
(2) 4.5.3 - Rated output, submerged
(3) 4.5.4 - Voltage regulation
(4) 4.5.5.1 - Voltage transient - (at 77°F for 1/2 hour)
(5) 4.5.5.3 - Ignition lead current
(6) 4.5.7 - Overspeed
(7) 4.5.8 - High temperature (for 20 hours)
(8) 4.5.9 - Low temperature
(9) 4.5.10 - Shock
(10) 4.5.11 - Vibration (for 1 hour along each axis)

NOTE: In addition to the above tests, the following sequence items (11) through (15) shall be performed at room temperature except where otherwise specified and while connected in a circuit as shown in figure 1. During this part of the test the voltage shall remain within the limits of set voltage ± 0.5 volts.

(11) 100 hours - 6000 RPM - 30 AMPS, in air.
(submersion - 15 minutes with maximum rated output)

(see 4.5.3, 6000 RPM)

(12) High temperature - (20 hours, See 4.5.8)
48 hours - 1750 ± 50 AMP - 30 AMPS in air

(13) Sand and dust - room ambient
Run 1 hour - 2000 RPM - No load, in air.

(14) Corrosion for 48 hours (condition B)
1 hour - 2000 RPM - No load, in air.

(15) 4.5.2 maximum rated output only.

4.6 Inspection of preparation for delivery.

4.6.1 Materials and processes. The Government inspector shall, at unscheduled intervals, inspect all materials and processes involved in the preparation for deli-

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very, to determine conformance to requirements of section 5 and specifications referenced herein. Any evidence of deviation from specified requirements shall be cause for refusal to conduct further inspection until objective evidence has been provided by the contractor that corrective action has been taken.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. Preservation, packaging, packing and marking shall be in accordance with the applicable packaging standard or packaging data sheet specified by the procuring activity (see 6.2).

6. NOTES

6.1 Intended use. Generator systems covered by this specification are intended for use in military motor vehicles, and on other military equipment to produce direct-current electrical energy for vehicular and various other requirements such as lighting and battery charging under extreme or unusual conditions of climate, weather terrain or military services.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Type and military part number.
(c) Federal stock number.
(d) Address to which preproduction samples shall be sent (see 4.2).
(e) Selection of applicable packaging standard or packaging date sheet (see 5.1).
(f) Desired voltage setting (see 3.8.2).

6.3 Vehicle loading. The following vehicle weights and loads are approximates (see 4.5.18):

Curb weight	Payload	Gross vehicle weight	Rated towed load
Highway	10,000 lbs	-	13,000 pounds
Cross-country	10,000 lbs	-	

6.3.1 Curb weight Curb weight included weight of vehicle with all attachments, accessories, OVM and equipment, full complement of fuel, lubricants, and coolant.

6.3.2 Payload. Vehicle payload shall include driver, co-driver, and 16 personnel or 5 tons of cargo.

6.3.3 Towed load. With towed load, the vehicle performance as specified herein shall be met with the M656 towing the rated towed load. Total weight of trailer and cargo shall be limited to the rated towed loads specified herein.

6.4 Definition.

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6.4.1 Set voltage. As used in this specification the term "set voltage" is deined as the output voltage to which the generator system is adjusted. The value of the voltage where referenced herein is in the range between 27.8 and 28.2 volts. After setting, the voltage reading shall be accurate to ± 0.075 volts.

Custodian:
Army - AT

Preparing activity:
Army - AT

Project No.
2920-A094

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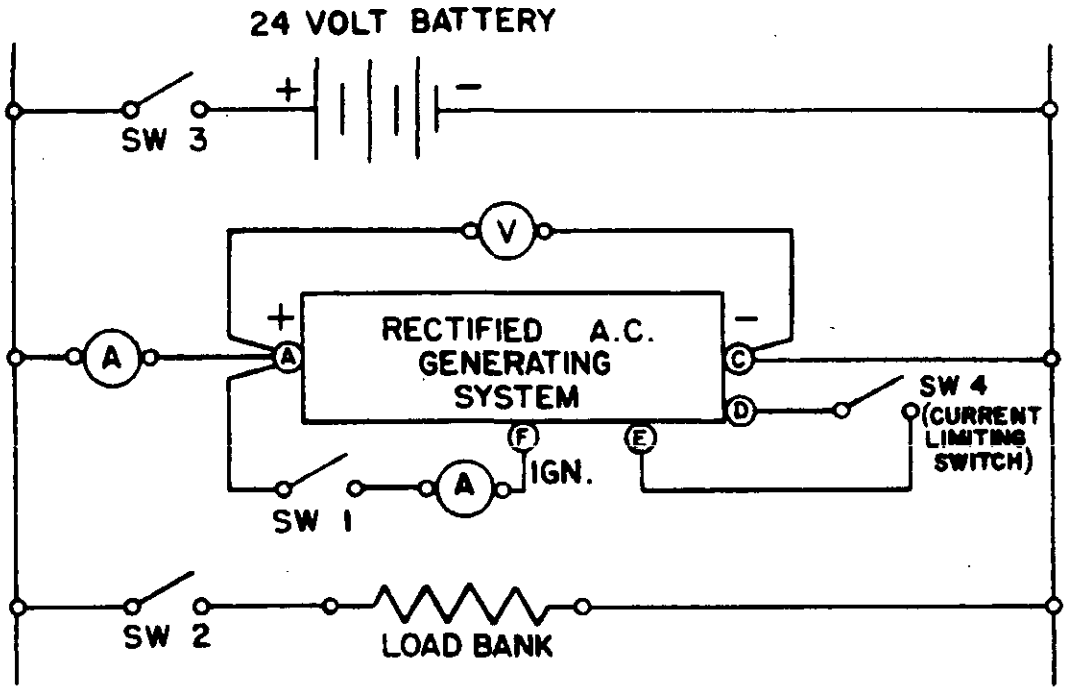


FIGURE 1 - GENERATOR SYSTEM CONNECTED IN TEST CIRCUIT

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